

HW 9

Townsend Problem 9.9 (5 pts)

The carbon monoxide molecule, CO, absorbs a photon with a frequency of 1.15×10^{11} Hz, making a purely rotational transition from an $l = 0$ to $l = 1$ energy level. What is the internuclear distance for this molecule?

The energy is exactly $\hbar\omega$ for the simple harmonic oscillator here.

The energy from $l = 0$ to $l = 1$

$$\Delta E = \frac{l\hbar^2}{I} = \frac{\hbar^2}{I}$$

Where

$$I = \mu r_0^2, \text{ so } r_0 = \sqrt{\frac{I}{\mu}}$$

We can set these equal and shove

$$I = \frac{\hbar^2}{\Delta E}$$
$$r_0 = \sqrt{\frac{\hbar^2}{\mu \Delta E}}$$

We can assume that μ is on the order of the nuclear mass M_n , but we can easily get μ here anyways

$$\mu = \frac{m_c m_o}{m_c + m_o}$$

This energy difference is

$$\Delta E = h\nu$$
$$\Delta E = 1.15 * 10^{11} * 6.626 * 10^{-34}$$
$$= 7.62 * 10^{-23} J$$

The mass of carbon is

$$\frac{12g}{6.02 * 10^{23}}$$
$$= 1.99 * 10^{-26} kg$$

Similarly, the mass of oxygen is

$$2.6 * 10^{-26} kg$$

this gives us

$$\mu = 1.09 * 10^{-26}$$
$$r_0 = \sqrt{\frac{\hbar^2}{\mu \Delta E}}$$
$$= 1.15 * 10^{-10} m$$
$$= 1.15 \text{\AA}$$

where $\text{\AA}$ is angstrom, i don't know the latex and have to rush to finish the pset lol.

Townsend Problem 9.13 (5 pts)

A particle is in the orbital angular momentum state $|l, m\rangle$. Evaluate ΔL_x and ΔL_y for this state. Which states satisfy the equality in the uncertainty relation

$$\Delta L_x \Delta L_y \geq \frac{\hbar}{2} |\langle L_z \rangle|$$

Suggestion: One approach is to use $\hat{L}_x = (\hat{L}_+ + \hat{L}_-)/2$, and so on. Another is to take advantage of the symmetry of the expectation values of L_x^2 and L_y^2 in an eigenstate of $\hat{L}_z$.

Note: This could have been a question about general angular momentum—nothing is specific here to orbital.

$$\hat{L}_x = \frac{\hat{L}_+ + \hat{L}_-}{2}$$

$$\hat{L}_y = \frac{\hat{L}_+ - \hat{L}_-}{2i}$$

The expectation value $\langle \hat{L}_x \rangle$ is

$$\begin{aligned} \langle lm | \hat{L}_x | lm \rangle &= \frac{1}{2} \langle lm | \hat{L}_+ + \hat{L}_- | lm \rangle \\ &= \frac{1}{2} \langle lm | \hat{L}_+ | lm \rangle + \frac{1}{2} \langle lm | \hat{L}_- | lm \rangle \\ &= 0 \end{aligned}$$

We can see this because raising the state changes the ket, but not the bra.

The same for $\langle \hat{L}_y \rangle = 0$

$$\Delta \hat{L}_x$$

We can now look at

$$\begin{aligned} \langle \hat{L}_x^2 \rangle &= \langle l, m | \hat{L}_x \hat{L}_x | lm \rangle \\ &= \frac{1}{4} \langle lm | (\hat{L}_+ + \hat{L}_-)(\hat{L}_+ + \hat{L}_-) | l, m \rangle \\ &= \frac{1}{4} \langle lm | \hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+ + \cancel{\hat{L}_+^2} + \cancel{\hat{L}_-^2} | l, m \rangle \end{aligned}$$

We know that

$$\hat{L}_- |l, m\rangle = \sqrt{l(l+1) - m(m-1)} \hbar |l, m-1\rangle$$

And that

$$\hat{L}_+ |l, m\rangle = \sqrt{l(l+1) - m(m+1)} \hbar |l, m+1\rangle$$

Therefore,

$$\begin{aligned} \langle lm | \hat{L}_+ \hat{L}_- | lm \rangle &= \sqrt{l(l+1) - m(m-1)} \hbar \langle lm | \hat{L}_+ | l, m-1 \rangle \\ &= \sqrt{l(l+1) - m(m-1)} \sqrt{l(l+1) - (m-1)m} \hbar^2 \langle lm || l, m \rangle \\ &= (l(l+1) - m(m-1)) \hbar^2 \end{aligned}$$

And

$$\begin{aligned}
& \langle l, m | \hat{L}_- \hat{L}_+ | lm \rangle \\
&= \sqrt{l(l+1) - m(m+1)} \hbar \langle l, m | \hat{L}_- | l, m+1 \rangle \\
&= \sqrt{l(l+1) - m(m+1)} \sqrt{l(l+1) - (m+1)m} \hbar^2 \\
&= (l(l+1) - m(m+1)) \hbar^2
\end{aligned}$$

The expectation value is therefore

$$\langle \hat{L}_x^2 \rangle = \hbar^2 \left(\frac{l(l+1) - m(m-1)}{4} + \frac{l(l+1) - m(m+1)}{4} \right)$$

The m bits combine

$$\begin{aligned}
& m(m-1) - m(m+1) \\
& m(m-1 - m-1) \\
& -2m
\end{aligned}$$

so we get

$$\langle \hat{L}_x^2 \rangle = \frac{2l(l+1) - 2m}{4}$$

This gives us

$$\Delta \hat{L}_x = \frac{2l(l+1) - 2m}{4}$$

$$\Delta \hat{L}_y$$

$$\begin{aligned}
\langle \hat{L}_x^2 \rangle &= \langle l, m | \hat{L}_x \hat{L}_x | lm \rangle \\
&= \frac{1}{4i^2} \langle lm | (\hat{L}_+ - \hat{L}_-)(\hat{L}_+ - \hat{L}_-) | l, m \rangle \\
&= \frac{1}{4} \langle lm | -\hat{L}_+ \hat{L}_- - \hat{L}_- \hat{L}_+ + \cancel{\hat{L}_+^2} + \cancel{\hat{L}_-^2}^0 | l, m \rangle \\
&= \frac{1}{4} \langle lm | \hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+ | l, m \rangle
\end{aligned}$$

We can see that this is the same as $\hat{L}_x^2$, so

$$\Delta \hat{L}_y = \frac{2l(l+1) - 2m}{4}$$

Townsend Problem 9.23 (5 pts)

The Hamiltonian for a three-dimensional system with *cylindrical symmetry* is given by

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mu} + V(\hat{\rho})$$

where $\rho = \sqrt{x^2 + y^2}$.

- Use symmetry arguments to establish that both $\hat{p}_z$, the generator of translations in the z direction, and $\hat{L}_z$, the generator of rotations about the z axis, commute with $\hat{H}$.
- Use the fact that $\hat{H}$, $\hat{p}_z$, and $\hat{L}_z$ have eigenstates in common to express the position-space eigenfunctions of the Hamiltonian in terms of those of $\hat{p}_z$ and $\hat{L}_z$. *Suggestion:* Follow a strategy similar to the one that we followed in (9.133) for a spherically symmetric potential except that here we are using the eigenfunctions of $\hat{p}_z$ and $\hat{L}_z$ instead of the eigenfunctions of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$.
- What is the radial equation? *Note:* The Laplacian in cylindrical coordinates is given by

$$\nabla^2 \psi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2}$$

We know that $\hat{L}_z$ commutes with $\hat{P}^2$. The potential only depends on x and y , so any actions on z don't matter which order they are done in. Therefore, the Hamiltonian commutes with $\hat{L}_z$.

B

For convenience, I'll label eigenstates of $\hat{H}$ as $|e\rangle$, eigenstates of $\hat{P}^2$ as $|P\rangle$, eigenstates of $\hat{L}_z$ as $|l\rangle$, and eigenstates of $\hat{p}_z$ as $|p\rangle$

We want to find

$$\langle e, l, m || \rangle$$

Townsend Problem 9.12 (5 pts)

The wave function for a particle is of the form $\psi(\mathbf{r}) = (x + y + z)f(r)$. What are the values that a measurement of L^2 can yield? What values can be obtained by measuring L_z ? What are the probabilities of obtaining these results? *Suggestion:* Express the wave function in spherical coordinates and then in terms of the Y_{lm} 's.

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} \\ x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned}$$

We can rewrite this in spherical

$$\begin{aligned} \psi(r) &= (x + y + z)f(r) = \sqrt{r}f(r) \\ &= (\sin \theta \cos \phi + \sin \theta \sin \phi + \cos \theta)r f(r) \end{aligned}$$

The $\cos \theta$ part can be represented by

$$Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos \theta$$

The $\sin \theta \sin \phi$ can be represented with

$$\begin{aligned} (Y_{11} + Y_{1,-1}) &= -\sqrt{\frac{3}{8\pi}} \sin \theta (e^{i\phi} - e^{-i\phi}) \\ &= -i\sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi \end{aligned}$$

the $\sin \theta \cos \phi$ is just

$$\begin{aligned} (Y_{1,-1} - Y_{11}) &= \sqrt{\frac{3}{8\pi}} \sin \theta (e^{i\phi} + e^{-i\phi}) \\ &= \sqrt{\frac{3}{4\pi}} \sin \theta \cos \phi \end{aligned}$$

Adding these just gives

$$\begin{aligned} \psi(r) &= \sqrt{\frac{4\pi}{3}} (Y_{1,-1} - Y_{1,1} + Y_{1,1} + Y_{1,-1} + Y_{1,0}) r f(r) \\ &= \sqrt{\frac{4\pi}{3}} (2Y_{1,-1} + Y_{1,0}) r f(r) \\ \hat{L}_z |l, m\rangle &= m\hbar |l, m\rangle \end{aligned}$$

When we measure $\hat{L}_z$, we can measure either $-\hbar \frac{2}{3}$ of the time, or $0, \frac{1}{3}$ of the time.

Townsend Problem 9.15 (5 pts)

Show that the spherical harmonics $Y_{l,m}$ are eigenfunctions of the parity operator with eigenvalue $(-1)^l$. *Note:* An inversion of coordinates in spherical coordinates is accomplished by $r \rightarrow r, \theta \rightarrow \pi - \theta$, and $\phi \rightarrow \phi + \pi$. Use (9.148). It may be wise to check the specific examples in (9.151) through (9.153) before attempting the general case (9.148).

We can act the parity operator on the state

$$Y_{lm} = \frac{(-1)^l}{2^l l!} \sqrt{\frac{(2l+1)(l+m)!}{4\pi(l-m)!}} e^{im\phi} \frac{1}{\sin^m \theta} \frac{d^{l-m}}{d(\cos \theta)^{l-m}} \sin^{2l} \theta$$

I will ignore all the constants and only look at how the parity operator acts on the coordinate filled portions of Y_{lm} , which I'll call y_{lm} for now

$$Y_{lm} \propto e^{im\phi} \frac{1}{\sin^m \theta} \frac{d^{l-m}}{d(\cos(\theta))^{l-m}} \sin^{2l}(\theta) = y_{lm}$$

$$\hat{\Pi} y_{l,m} = e^{im(\phi+\pi)} \frac{1}{\sin^m(\pi-\theta)} \frac{d}{d(\cos(\pi-\theta))^{l-m}} \sin^{2l}(\pi-\theta)$$

We know that

$$e^{im\pi} = (-1)^m$$

and

$$\begin{aligned} \sin(\pi - \theta) &= -\sin(\theta) \\ \cos(\pi - \theta) &= -\cos(\theta) \end{aligned}$$

This becomes

$$(-1)^m \frac{1}{(-1)^m \sin^m(\theta)} \frac{d}{d((-1)^{l-m}(\cos \theta)^{l-m})} \sin^{2l}(\theta)$$

This has just put a constant onto the original state y_{lm} ,

$$\begin{aligned} \lambda &= \frac{(-1)^m}{(-1)^m (-1)^{l-m}} \\ \lambda &= (-1)^{m-l} \end{aligned}$$

This is different from the expected result, but I don't know how to deal with the derivative with respect to $\cos$, and I'm unfortunately low on time before this is due.

Townsend Problem 9.20 (5 pts)

The wave function of a rigid rotator with a Hamiltonian $\hat{H} = \hat{L}^2/2I$ is given by

$$\langle \theta, \phi | \psi(0) \rangle = \sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi$$

- What is $\langle \theta, \phi | \psi(t) \rangle$? *Suggestion:* Express the wave function in terms of the $Y_{l,m}$'s and the state in terms of $|\ell, m\rangle$ kets.
- What values of L_z will be obtained if a measurement is carried out and with what probability will these values occur?
- What is $\langle L_x \rangle$ for this state? *Suggestion:* Use bra-ket notation and express the operator $\hat{L}_x$ in terms of raising and lowering operators.
- If a measurement of L_x is carried out, what result(s) will be obtained? With what probability? *Suggestion:* If you have worked out Problem 3.15, you can take good advantage of the expressions for the states $|1, m\rangle_x$. Specifically, take advantage of symmetry and

$$|1, 0\rangle_x = \frac{1}{\sqrt{2}}|1, 1\rangle - \frac{1}{\sqrt{2}}|1, -1\rangle$$

a

We can break this up into

$$Y_{11} = -\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi} |1, 1\rangle$$

$$Y_{1,-1} = \sqrt{\frac{3}{8\pi}} \sin \theta e^{-i\phi} |1, -1\rangle$$

The sum of these is

$$(Y_{11} + Y_{1,-1}) = -\sqrt{\frac{3}{8\pi}} \sin \theta (e^{i\phi} - e^{-i\phi})$$

$$= -i\sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi$$

Therefore,

$$\frac{1}{i}(Y_{11} - Y_{1,-1}) = \langle \theta, \phi | \psi(0) \rangle$$

The spherical harmonics are energy eigenstates of the Hamiltonian and have the same E

$$Y_{l,m}(t) = e^{-i\frac{E}{\hbar}t} Y_{l,m}(0)$$

Therefore, the time evolved ψ has every component multiplied by the same spinning phase

$$\psi(t) = e^{-i\frac{E}{\hbar}t} \psi(0)$$

b

$$\hat{L}_z |l, m\rangle = m\hbar |l, m\rangle$$

We have an equal mixture of $Y_{1,1}$ and $Y_{1,-1}$. This gives a 50-50 chance of either $\hbar$ or $-\hbar$ angular momentum.

c

$$\hat{J}_x = \frac{\hat{J}_+ + \hat{J}_-}{2}$$

We can apply the operator and do

$$\langle lm | \hat{J}_x | lm \rangle$$

We can't raise the $Y_{1,1}$ state, nor can we lower the $Y_{1,-1}$ state - this just leaves us with what we started - the lowered Y_{11} and the raised $Y_{1,-1}$. This is exactly what we started with, but with a factor of $\frac{1}{2}$.

The eigenvalues of $\hat{J}_+$ are $\sqrt{j(j+1) - m(m+1)}\hbar$ and $\hat{J}_-$ are $\sqrt{j(j+1) - m(m-1)}\hbar$.

For the $Y_{1,-1}$ state being raised, this gives

$$\sqrt{1(2) - (-1)(1)} = \sqrt{2+1} = \sqrt{3}$$

For $Y_{1,1}$ being lowered, this is

$$\sqrt{1(2) - (1)(-1)} = \sqrt{3}$$

We get $2\sqrt{3}$ from these alone, but we have the factor of $\frac{1}{2}$.

Therefore, $\hat{L}_x = \frac{1}{2}(2)\hbar = \sqrt{3}\hbar$

d

$$\underline{|1,0\rangle_x = \frac{1}{\sqrt{2}}|1,1\rangle - \frac{1}{\sqrt{2}}|1,-1\rangle}$$